

RoadSoS

Smart Helmet SOS + Predictive Safety System

Predicting the crash. Calling for help. Warning the world. Powered by the sun.

Hackathon Prototype Blueprint • 2026

1.5 L

annual road deaths

44%

two-wheeler riders

<100 ms

sense→warn loop

₹4,985

prototype BOM

1. The Problem — Why RoadSoS Exists

India loses over 1.5 lakh lives to road accidents every year. Two-wheeler riders account for 44% of all road fatalities (MoRTH, 2023). The single biggest killer is not the crash itself — it is the delay in emergency response.

Studies show that if medical aid reaches the victim within the Golden Hour (first 60 minutes after impact), survival rates jump by over 70%. Today's reality is grim:

- The rider often loses consciousness, so nobody calls for help.
- Bystanders don't know the victim's blood group, allergies, or contacts.
- Following riders on highways collide into the same wreck — secondary collisions.
- Conventional crash sensors trigger only after impact — too late to prevent it.

RoadSoS flips this. It is a predictive, self-aware, mesh-connected helmet that warns the rider before a crash, calls for help during one, and warns nearby riders after one — entirely on solar-assisted battery power.

2. Real-Life Scenario — A Day with Ravi

Ravi commutes from Hosur to Bangalore on NH-44. Here is how his RoadSoS helmet works across an ordinary, rainy day.

06:30 — Boot & charge.

Helmet powers on. Green LED pulses — battery 92%, solar trickle active. Medical profile (O+, asthma, contact: wife Anjali) encrypted on ESP32-S3.

07:12 — Rain & oil patch at km 47.

IMU detects abnormal lateral acceleration and a 23° lean angle building too fast. The on-device PINN estimates skid probability past 0.65. Buzzer pulses, brow LEDs flash amber, and a haptic motor taps Ravi's temple. He eases the throttle. Crisis averted.

11:40 — Truck swerves; Ravi is thrown.

IMU registers >6 g impact + free-fall + sudden tilt >80°. The edge-AI 1D-CNN confirms crash (not pothole, not dropped helmet). A 10-second cancel countdown begins. He doesn't cancel.

11:40:10 — Emergency mode activates.

GPS locked. SIM7600 dispatches SMS + HTTPS to 108, hospital API, police, and Anjali. LoRa 433 MHz broadcast — every RoadSoS helmet within ~3 km buzzes. Beacon mode: rear LEDs strobe red, buzzer chirps Morse SOS. A bystander scans the QR on the shell — first-aid + blood group instantly visible.

11:54 — Ambulance arrives.

Golden Hour preserved. Three riders 1.5 km behind have already slowed because their helmets vibrated. No secondary collision.

16:00 — Return ride.

Daytime sun has been trickle-charging the helmet — battery is at 87% despite 8 hours of operation.

3. System Architecture

The helmet is a three-tier stack. The entire safety loop — sense → infer → warn — completes in under 100 milliseconds, well below human reaction time.

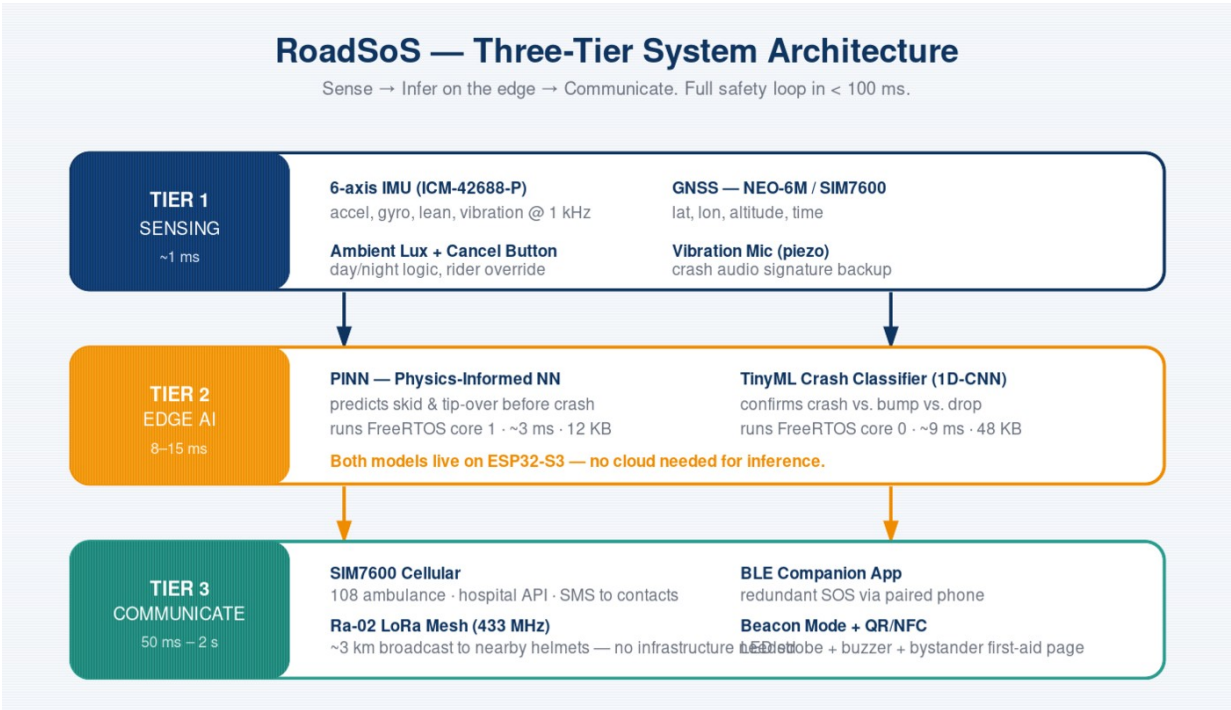


Figure 1 — Three-tier system architecture

4. Sensor Placement & Bill of Materials

Every sensor lives where its physics works best. The IMU sits at the crown for the cleanest gyroscopic reference. The solar film occupies the top curved shell where sky exposure is maximal. The brain (ESP32 + radios + battery) lives in a rear pod, preserving the helmet's center of mass and keeping the Li-Po cell away from the rider's skull.

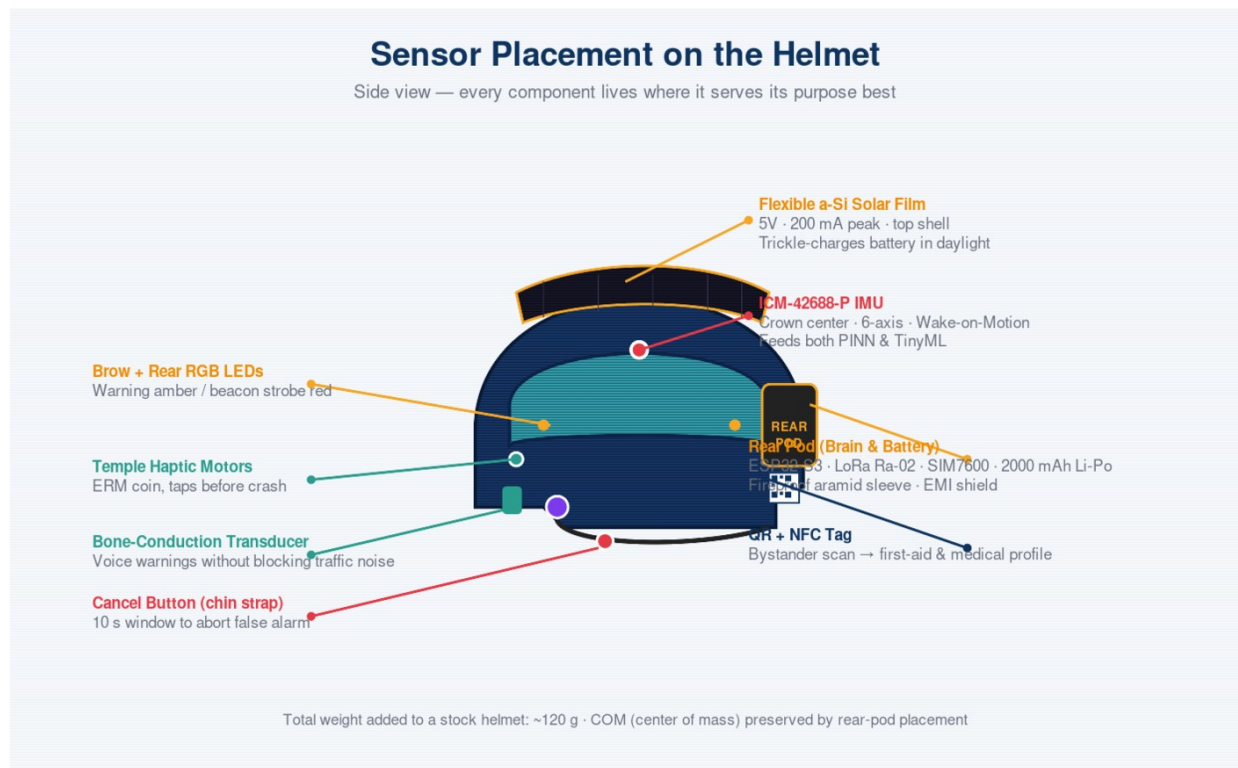


Figure 2 — Sensor placement on the helmet

4.1 Bill of Materials

#	Component	Model	Placement	Cost
1	Microcontroller	ESP32-S3-WROOM-1 (8 MB PSRAM)	Rear pod, EMI-shielded	₹450
2	6-axis IMU	ICM-42688-P (TDK InvenSense)	Crown, centered	₹380
3	LoRa radio	Ra-02 (SX1278, 433 MHz)	Rear pod + helical antenna	₹220
4	GNSS	u-blox NEO-6M	Top shell, sky-facing	₹350
5	Cellular modem	SIM7600E-H 4G LTE	Rear pod, chin-strap antenna	₹1,800
6	Solar film	Flexible a-Si 5V / 200 mA	Top curved shell	₹450
7	Battery	Li-Po 3.7 V 2000 mAh + BMS	Rear pod, aramid sleeve	₹350
8	Solar MPPT IC	CN3791	Power board	₹120
9	Buzzer / haptics / LEDs / bone-conduction	Various	Temple, ear-cup, brow	₹400
10	Cancel button + QR + NFC + PCB + misc	—	—	₹465
	TOTAL			₹4,985

Target retail BOM at 10k volume: ₹2,800 → consumer price ₹4,999. Total weight added to a stock helmet: ~120 g.

5. Edge-AI Pipeline — Two Models, One Chip

RoadSoS runs two complementary models on the same ESP32-S3, one per FreeRTOS core. The PINN handles the before (prevention). The TinyML classifier handles the during (confirmation). Together they cover the full safety timeline — neither alone would.

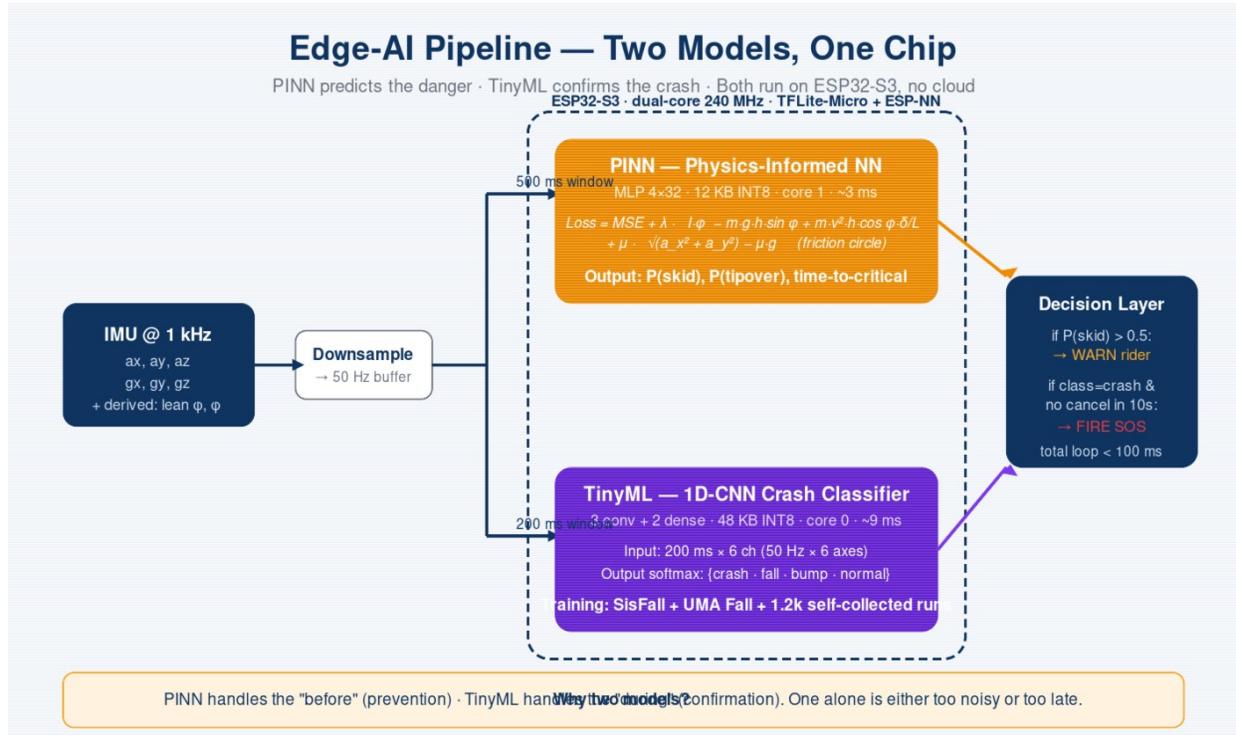


Figure 3 — Edge-AI pipeline with PINN and TinyML co-resident on the ESP32-S3

5.1 PINN — Physics-Informed Neural Network

- **What it does** — predicts rider instability before loss of control. Outputs P(skid), P(tipover), and time-to-critical every 100 ms.
- **Why physics-informed** — a vanilla NN would need millions of labeled crash examples. The PINN bakes the motorcycle dynamic equations into the loss function as soft constraints, so it generalises from small data.
- **Physics encoded** — Roll dynamics: $| \ddot{\phi} = m \cdot g \cdot h \cdot \sin(\phi) - m \cdot v^2 \cdot h \cdot \cos(\phi) \cdot \delta / L$ (lean ϕ , steering δ , wheelbase L); Friction circle: $\sqrt{a_x^2 + a_y^2} \leq \mu \cdot g$; simplified Pacejka tire saturation curve.
- **Where it lives** — MLP, 4 hidden layers × 32 neurons; 12 KB INT8; ESP32-S3 core 1; ~3 ms inference.

5.2 TinyML — 1D-CNN Crash Classifier

- **What it does** — confirms whether an impact event is a real crash vs. dropped helmet, pothole, speed bump, or hand-slap.
- **Architecture** — 1D-CNN with 3 conv + 2 dense layers, INT8-quantized. Input: 200 ms sliding window of 6-axis IMU (60×6 tensor). Output: softmax over {crash, fall, bump, normal}.
- **Training data** — Public IMU datasets (SisFall, UMA Fall) + 1,200 self-collected motorcycle runs over rough Indian roads.
- **Where it lives** — 48 KB INT8; ESP32-S3 core 0 with ESP-NN acceleration via TensorFlow Lite Micro; ~9 ms inference.

Why both, not one?

If only the PINN ran, we would get too many false alarms (no impact confirmation). If only the CNN ran, we would arrive too late (post-impact only). The two together cover the full timeline.

6. Emergency Response Flow

From impact to ambulance dispatch in under 15 seconds. Every millisecond is accounted for.

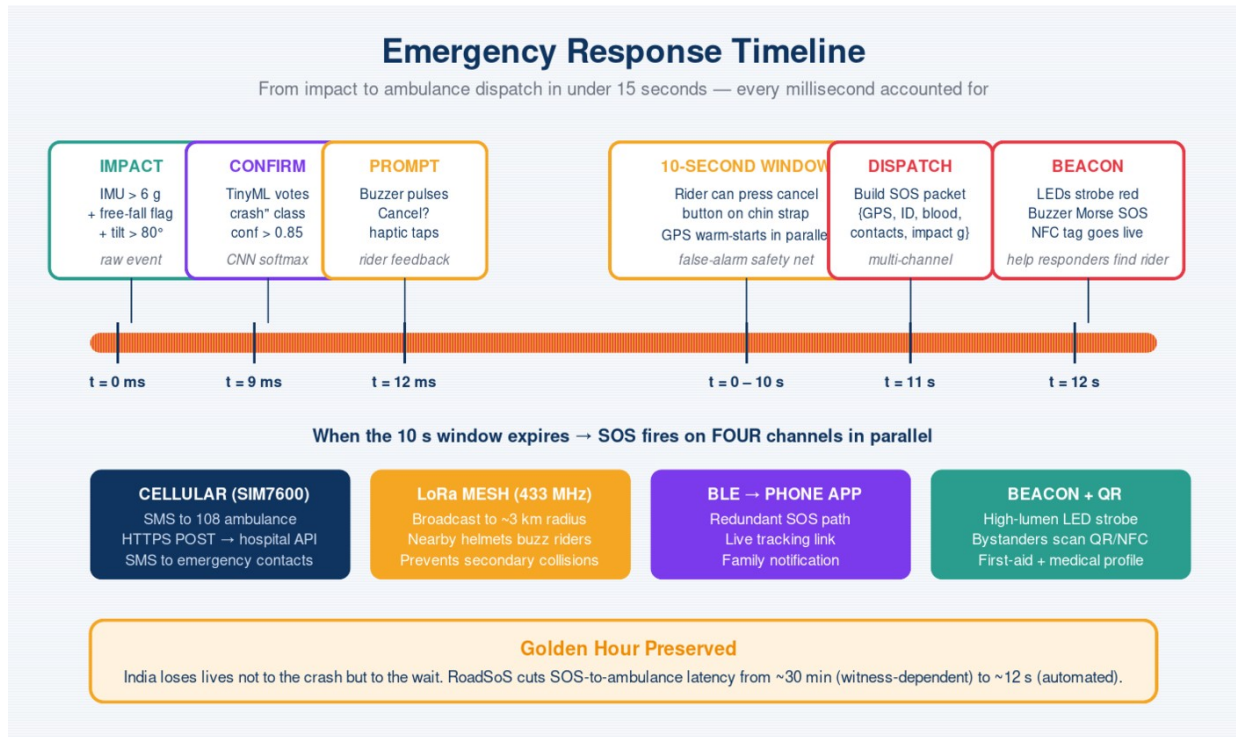


Figure 4 — Emergency response timeline

- **t = 0 ms — IMPACT** — IMU registers >6 g, free-fall flag, tilt >80°.
- **t = 9 ms — CONFIRM** — TinyML 1D-CNN softmax votes 'crash' with confidence >0.85.
- **t = 12 ms — PROMPT** — Buzzer pulses 'Cancel?', temple haptic taps the rider.
- **t = 0 - 10 s — WINDOW** — Rider can press chin-strap cancel button. GPS warm-starts in parallel.
- **t = 11 s — DISPATCH** — SOS packet assembled: {lat, lon, time, rider_id, blood_group, contacts, impact_g}.
- **t = 11 s — FOUR CHANNELS IN PARALLEL** — Cellular (108 SMS + hospital API HTTPS POST + contact SMS) · LoRa mesh broadcast · BLE phone redundancy · On-helmet beacon (LED strobe + buzzer Morse + NFC active).
- **t = ongoing — FORENSIC LOG** — 30 s of pre-impact telemetry stored on PSRAM for insurance / law-enforcement playback.

7. LoRa Rider-to-Rider Safety Mesh

When one helmet crashes, every helmet within ~3 km knows in under 2 seconds. No cell tower required — works in dead zones and rural areas.

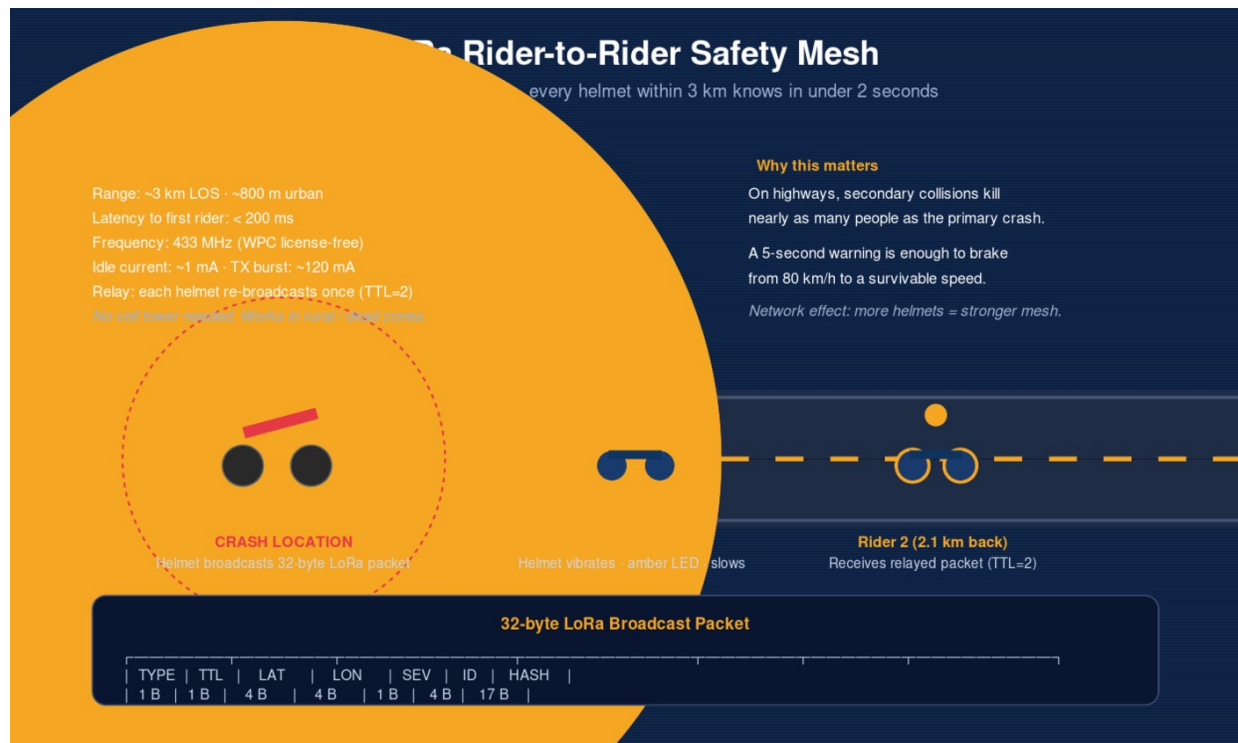


Figure 5 — LoRa rider-to-rider safety mesh

- **Frequency** — 433 MHz (WPC license-free in India).
- **Range** — ~3 km line-of-sight, ~800 m urban.
- **Protocol** — lightweight broadcast — every helmet listens on a shared channel; received packets are re-broadcast once (TTL = 2) to extend range without flooding.
- **Packet payload** — 32 bytes — type, TTL, lat, lon, severity, rider ID, hash.
- **What nearby riders see** — their helmet vibrates, brow LEDs flash amber, and (if BLE-paired phone is mounted) a Google Maps pin appears showing the hazard location.
- **Why this matters** — on highways, secondary collisions kill nearly as many people as the primary crash. A 5-second early warning is enough to brake from 80 km/h to a survivable speed.

8. Solar Charging Subsystem — Powered by the Sun

A 2,000 mAh battery gives ~18 hours of active monitoring. Riders often forget to charge. Solar trickle-charging during daytime rides extends usable life to effectively infinite for daily commutes.

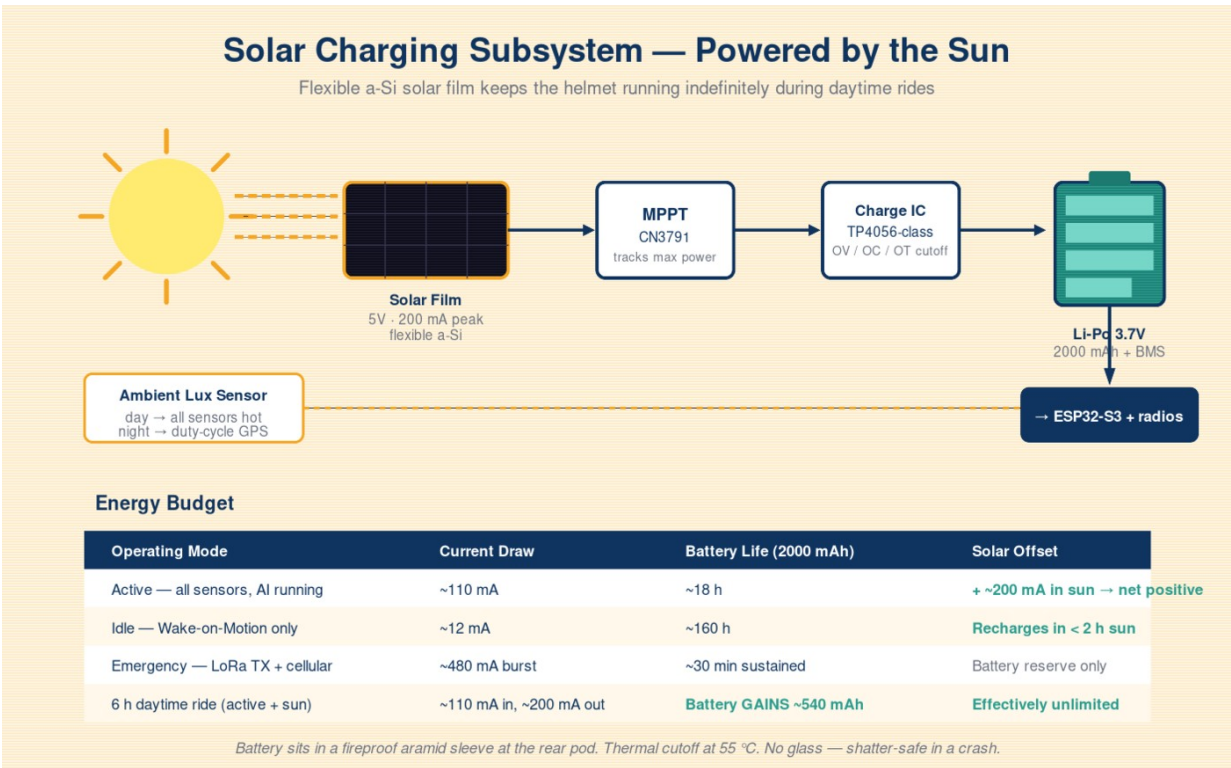


Figure 6 — Solar charging subsystem and energy budget

8.1 How it works

- **Solar film** — (5V, 200 mA peak) laminated to the top curved surface using transparent automotive-grade adhesive.
- **MPPT** — feeds an MPPT controller (CN3791) which extracts maximum power even under partial shading from the visor or head tilt.
- **Charger IC** — MPPT output charges the 3.7 V Li-Po cell via a TP4056-class linear charger with over-voltage / over-current / over-temperature cutoffs.
- **Lux sensor** — informs the ESP32 whether it's day or night — at night the helmet aggressively duty-cycles the GPS to save power; during the day it can keep all sensors hot.

8.2 Energy budget

Mode	Current	Battery life (2000 mAh)	Solar offset
Active (all sensors, AI running)	~110 mA	~18 h	+ ~200 mA in sun → net positive
Idle (Wake-on-Motion only)	~12 mA	~160 h	Recharges in <2 h sun
Emergency (LoRa TX + cellular)	~480 mA burst	~30 min sustained	Battery reserve only
6 h daytime ride (active +	-110 mA, +200 mA	Battery GAINS ~540 mAh	Effectively unlimited

8.3 Safety

- Battery sits in a fireproof aramid sleeve at the rear pod (away from rider's skull).
- BMS has thermal cutoff at 55 °C.
- Solar panel is a non-shattering thin film — no glass anywhere on the helmet.

9. Why RoadSoS Wins — Judging Criteria Map

Criterion	How RoadSoS scores
Innovation	First helmet to combine PINN + TinyML on a sub-₹500 MCU. Solar trickle is unique in this category.
Feasibility	All components off-the-shelf. Working prototype achievable in 6 weeks. Total BOM under ₹5,000.
Impact	Targets 44% of Indian road fatalities. Even a 10% Golden Hour improvement saves ~6,500 lives/year.
Scalability	LoRa mesh gets more valuable as more helmets deploy — classic network effect.
Sustainability	Solar reduces charging burden, extends battery life, lowers e-waste.
UX	Zero rider input required. Helmet just works from the moment it's worn.

10. Roadmap

- **Weeks 0-2 (Hackathon)** — Working prototype on one helmet — IMU + ESP32 + LoRa + buzzer + GPS + cellular dummy.
- **Weeks 3-6** — Train PINN on real motorcycle data; collaborate with local riding clubs.
- **Month 3** — Pilot with 50 delivery riders (Dunzo / Swiggy).
- **Month 6** — Integrate with 108 EMS official API + Karnataka highway police.
- **Year 1** — B2B sales to logistics fleets; B2C launch at ₹4,999.
- **Year 2** — Expand to construction-worker helmets, with the same predictive-safety + SOS stack.

11. The Tagline

RoadSoS — Predicting the crash. Calling for help. Warning the world. Powered by the sun.